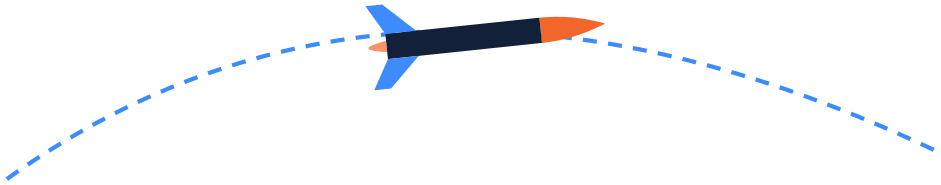
A 2-WEEK HOME-SCHOOL UNIT · GRADES 3–8

Rockets & Model Rocketry

Worksheets, data sheets, the inclinometer template and launch logs. Goes with **Rocket Builder by Quiver Design Works**, a free app at rocket.quiverdesignworks.net.



	Page	Use on	Who
W1	Balloon rocket data	Day 1	All ages
W2	Forces and launch angles	Day 2	All ages
W3	Swing test	Day 3	All ages
W4	Drag test	Day 4	All ages
W5	Motor codes and impulse	Day 5	All ages
W6	Design log	Day 6	All ages
W7	Parachute drop test	Day 7	All ages
W8	Altitude tracking	Day 8	All ages
T	Inclinometer template	Day 8	All ages
W9	Safety rules	Day 9	All ages
W9b	Graph your week	Day 9	All ages
W9c	Launch checklists	Day 9	All ages
L	Paper-rocket launch log	Day 10	All ages
F	Flight log (printed rocket)	Day 10	12+ only
C	Capstone planner and rubric	Day 10	All ages
Q	Quick checks, Days 1–10	Any day	All ages
K	Answer keys	Grown-ups	Grown-ups

Printing: US Letter, portrait, scale 100% ("actual size"), so the inclinometer comes out the right size. Turn on "background graphics" if you want the coloured bands. Print the answer keys only for the grown-up.

Two tracks: under 12 (about grades 3–6) flies paper-style rockets only (balloon, straw, paper and stomp rockets; fizzy rockets with an adult) and uses Rocket Builder on screen. **12+** also designs, prints and flies a Rocket Builder rocket on a real motor. Every page is for all ages unless it says **12+ only**.

Safety: model rocket motors are rated 12+, and grown-ups handle all motors and igniters. Wear safety glasses for straw, stomp and fizzy rockets. Follow the NAR Model Rocket Safety Code.

W1 Balloon rocket data

All ages

Day 1 · Newton's three laws

Words: force · mass · inertia · acceleration · action and reaction · thrust

Question: Does more air make it go further? Does more weight make it go less far?

My guess: _____

Part A: more air (same mass)

Balloon size	Circumference (cm)	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)
Small					
Medium					
Large					

Part B: more mass (same air)

Paper clips	Circumference (cm)	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)
0					
5					
10					

Average = (trial 1 + trial 2 + trial 3) ÷ 3. Grades 3–4: circle the longest trial instead.

Think about it

1. Newton's **third** law: the balloon pushes air out the back. What pushes the balloon forward?

2. Newton's **second** law: in Part B, the push was the same. What happened when you added paper clips? Why?

3. Newton's **first** law: why did the straw stay still until you let go? Why did it stop at the end?

4. Rocket Builder: press ▶ **Watch it launch**. Write three things you noticed.

Grades 7–8: a balloon pushes with 0.1 N on a 10 g (0.010 kg) rocket. Ignoring friction, a = F ÷ m = _____ m/s². With 10 g of paper clips added: a = _____ m/s².

 Name _____ Date _____

W2 Forces and launch angles

All ages

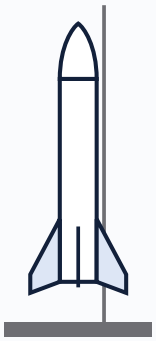
Day 2 · The four forces

Words: thrust · weight · drag · lift · balanced · unbalanced · apogee

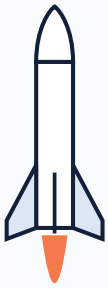
Part 1: draw the forces

Draw an arrow for each force on the rocket and label it: **thrust**, **weight**, **drag**. Make bigger forces longer arrows.

1. On the pad



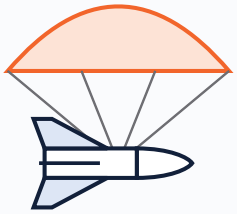
2. Boost (motor on)



3. Coast (motor off)



4. Under the parachute

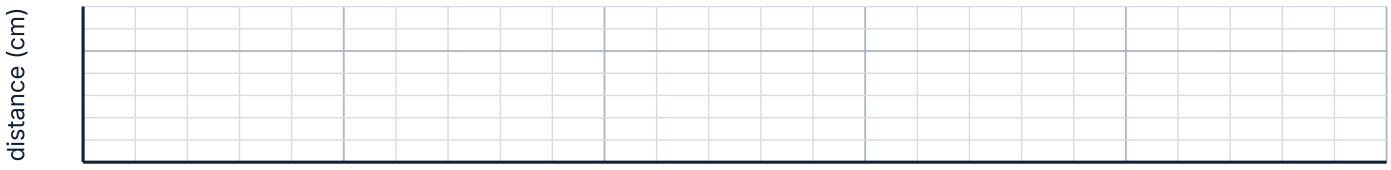


Which pictures show **balanced** forces? _____ Which show **unbalanced**? _____

Part 2: launch-angle test (straw rockets)

Angle	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)
15°				
30°				
45°				
60°				
75°				

Bar graph: _____



1. Which angle went furthest? Why do you think so?

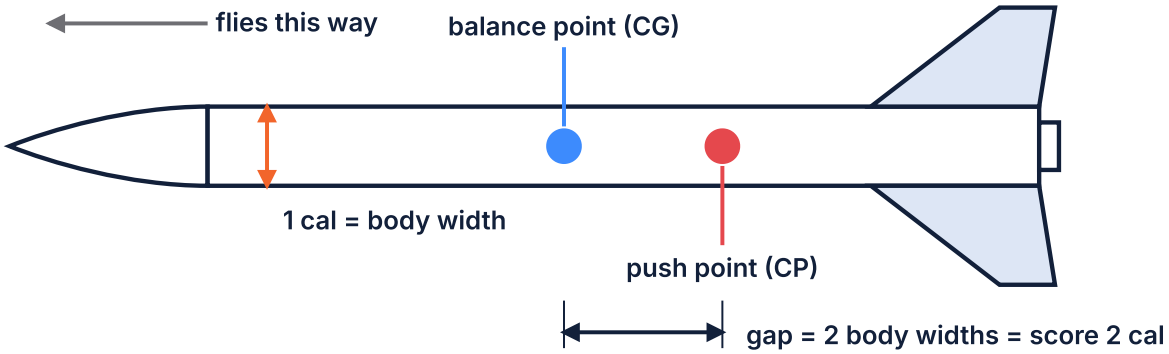
2. Rocket Builder: press ▶ **Watch it launch**. Number these in the order you see them: ___ coast ___ liftoff ___ parachute ___ top (apogee) ___ boost
3. **12+:** the **power** bar (thrust ÷ weight) for my rocket says _____ ×. Is it 5× or more? _____

W3 Swing test

All ages

Day 3 · Stability: balance point and push point

Words: stable · unstable · balance point (CG) · push point (CP) · caliber · stability score



The blue dot (balance point) must be **in front of** the red dot (push point). Score (cal) = (push point distance − balance point distance) ÷ body width. 1–2 cal is just right.

Body width of my test rocket: _____ cm (= 1 cal)

Version	Change	Balance point from tip (cm)	Push point from tip (cm)	Gap (cm)	Score (cal)	Swing test: stable, wobbly or tumbles?
A	no fins, no clay					
B	fins, no clay					
C	fins + clay in nose					
D	fins + clay at tail					

1. Which version was the most stable? Use your scores to explain.

2. Why did clay in the nose help, but clay at the tail hurt?

3. A rocket is 3 cm wide. Its push point is 4.5 cm behind its balance point. Score = _____ cal. Is that just right, wobbly, or too stable? _____
4. In Rocket Builder, my rocket's score is _____ cal. The fins are _____ shaped.

W4 Drag test

All ages

Day 4 · Drag and shape

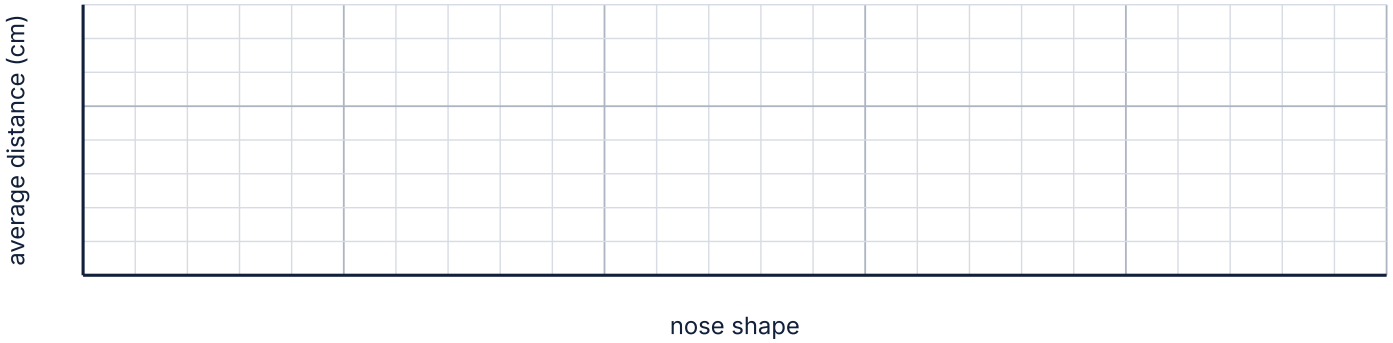
Words: drag · streamlined · frontal area · drag coefficient · variable · fair test · trade-off

The variable I'm changing: nose shape. **Things I keep the same:** _____

Test 1: nose shapes (launch angle: _____°)

Nose	Mass (g, optional)	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)
Flat					
Cone					
Round					

Bar graph: _____



Test 2: fins

Fins	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)	Straight or wobbly?
None					
3 small					
3 medium					
4 large					

1. Which nose had the least drag? How do you know?

2. Was the difference big, or could it be luck? What would make you more sure?

3. Explain the fin trade-off using your data.

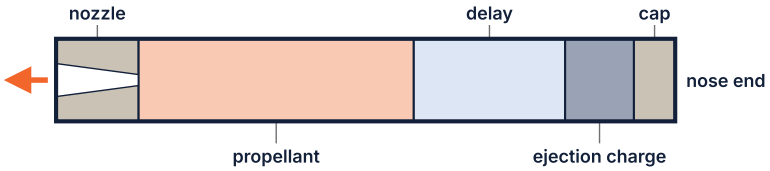
Grades 7–8: drag = ½ × 1.2 × speed² × Cd × area. A rocket with area 0.00071 m² and Cd 0.65 at 40 m/s: drag ≈ _____ N. In Rocket Builder (Advanced → Flight → Advanced), predicted altitude with Cd 0.5: _____ m; Cd 0.8: _____ m.

W5 Motor codes and impulse

All ages

Day 5 · Motors, thrust and impulse

Words: propellant · nozzle · delay · ejection charge · igniter · total impulse · impulse class · thrust-to-weight



hot gas rushes out of the nozzle, so the rocket is pushed the other way: toward the nose

Motor rockets are 12+, and only grown-ups handle motors and igniters. This drawing shows the inside of a motor. Never open one.

Class	1/4A	1/2A	A	B	C	D	E	F
Total impulse up to (N·s)	0.625	1.25	2.5	5	10	20	40	80
Smallest field (NAR)	15 m	15 m	30 m	60 m	120 m	150 m	300 m	300 m

1. Decode the motor codes

Code	Impulse class	Average thrust (N)	Delay (s)	Smallest field
A8-3				
B6-4				
C6-5				
D12-5				

2. Impulse = average thrust × burn time

Motor	Average thrust (N)	Burn time (s)	Total impulse (N·s)	Class
a	3	0.8		
b	5	0.9		
c	10	1.7		
d	15	3.4		

3. Rocket Builder and fizzy rockets (everyone)

The best motor for my design is _____ and it should fly about _____ m high. Fizzy rockets: the amount of water that worked best was _____ tsp.

4. Thrust-to-weight (grades 5–8; it must be 5 or more)

100 g weighs about 1 N. So a rocket of 80 g weighs about 0.8 N.

1. Rocket + motor = 80 g. Motor average thrust = 5 N. Weight = _____ N. Thrust ÷ weight = _____. Pass? _____
2. Rocket + motor = 120 g. Motor average thrust = 4.7 N. Weight = _____ N. Thrust ÷ weight = _____. Pass? _____

W5b Thrust curve

Grades 7–8

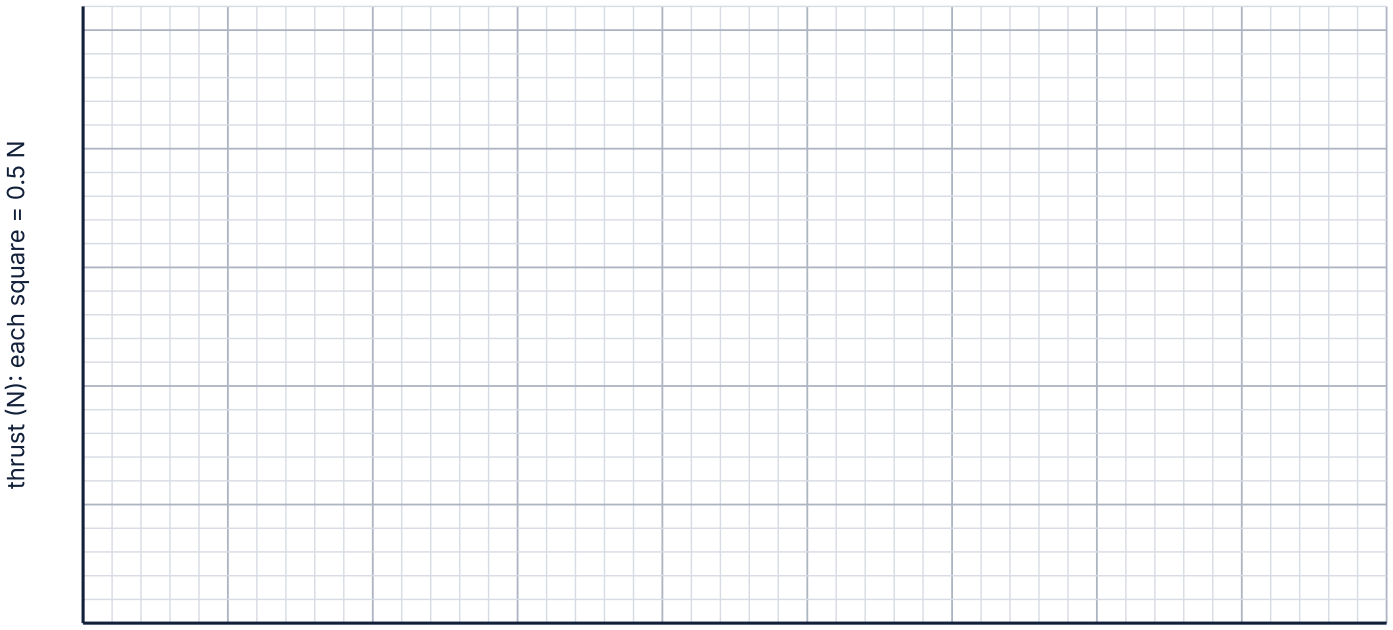
Day 5 · Grade 7–8 extension: impulse is the area under the thrust curve

This table is a **thrust curve** for a B motor, rounded from a measured one. Thrust is how hard the motor pushes at each moment.

Time (s)	0	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.85	0.9
Thrust (N)	0	12	9.5	5	4	4	4	4	4	4	4	0

1. Plot the points on the grid and join them with straight lines. Time along the bottom (each small square = 0.02 s), thrust up the side (each small square = 0.5 N).

Thrust curve, B motor _____



2. What is the **peak** (highest) thrust? _____ N. When? _____ s
3. The **area under the curve** is the total impulse. Find it by splitting the shape into triangles and rectangles (or count squares: each small square = 0.02 × 0.5 = 0.01 N·s). Total impulse ≈ _____ N·s. Class: _____
4. Average thrust = total impulse ÷ burn time (0.86 s) = _____ N. Why is the code's middle number a little different?

---

5. **F = m × a at liftoff.** Rocket + C6 motor: mass 0.086 kg, measured average thrust 4.7 N. Weight = 0.086 × 9.8 = _____ N. Net force = _____ N. Acceleration = _____ m/s².

W6 Design log

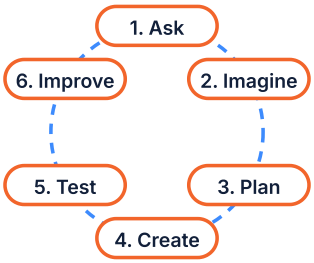
All ages

Day 6 · The engineering design cycle with Rocket Builder

Words: criteria · constraints · iteration · prototype · trend · model

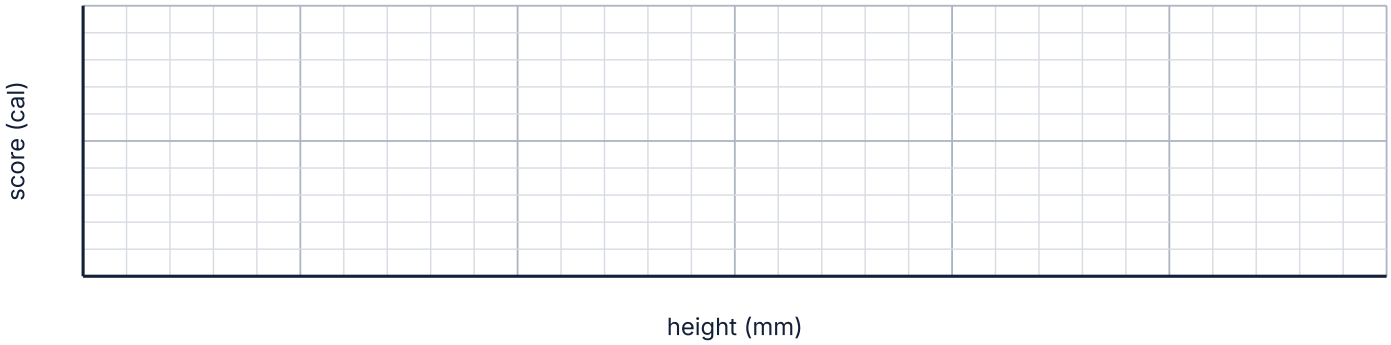
Criteria (what it must do): score 1–2 cal (2–3 OK) · a motor gets ✓ Good · height fits our field · _____

Constraints (limits): printer height _____ mm · field size _____ m · _____



#	Height (mm)	Fin shape	Nose	Score (cal)	Fins grew? ✓	Best motor	Altitude (m)	Power (×)	Off rod (m/s)
1									
2									
3									
4									
5									
6									
7									
8									

Line graph: height vs score _____



The trend: as the height went up, the score _____

Decision table: score each design 1 (poor) to 3 (great)

Design #	Stability (1–2 cal is best)	Altitude fits our field	Looks	Print time (12+)	Total

I chose design # _____ because _____

▶ **Watch it launch:** it went up about _____ m and landed about _____ m from the pad.

W7 Parachute drop test

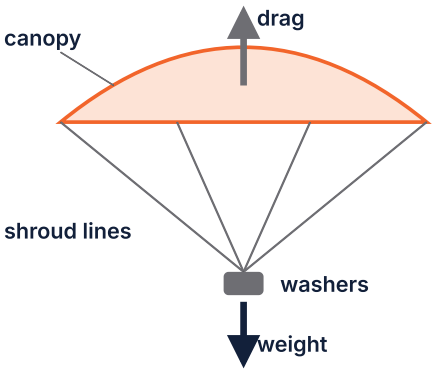
All ages

Day 7 · Parachutes and descent rate

Words: recovery · canopy · shroud lines · descent rate · terminal speed · drift

Drop height: _____ m **Weight on each parachute:** _____

Descent rate (m/s) = drop height ÷ average time

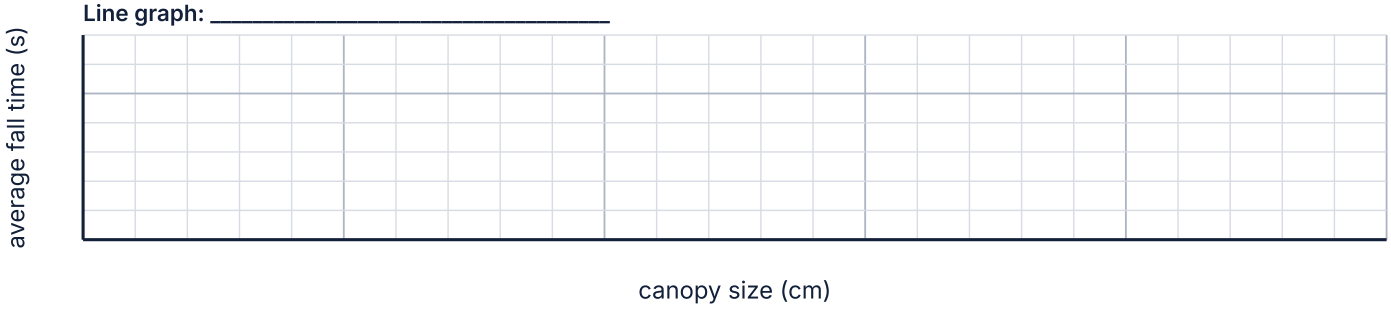


Parachute	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Average time (s)	Descent rate (m/s)
No parachute					
20 cm					
30 cm					
40 cm					

Part B (optional), 30 cm parachute:

Washers	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Average time (s)	Descent rate (m/s)
1					
2					
4					

Line graph: _____



Your rocket's parachute (Rocket Builder)

Parachute size: _____ inches. Descent speed: _____ m/s. Predicted altitude: _____ m.

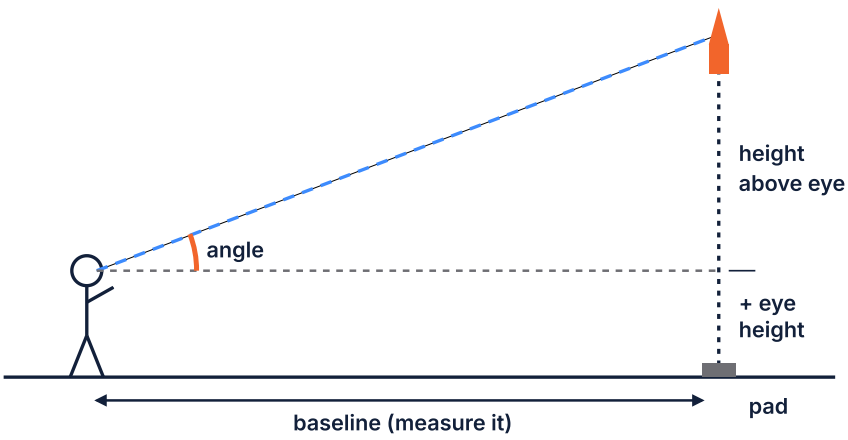
Time to come down = altitude ÷ descent speed = _____ s. Drift in a 10 mph (4.5 m/s) breeze = time × 4.5 = _____ m.

1. Practice: a rocket reaches 120 m and falls at 4 m/s. Time = _____ s. Drift = _____ m. Is a 60 m field big enough? _____
2. Why not just use the biggest parachute you can find?

W8 Altitude tracking

All ages

Day 8 · Measuring height with angles



Angle	Multiplier	Angle	Multiplier
5°	0.09	45°	1.00
10°	0.18	50°	1.19
15°	0.27	55°	1.43
20°	0.36	60°	1.73
25°	0.47	65°	2.14
30°	0.58	70°	2.75
35°	0.70	75°	3.73
40°	0.84	80°	5.67

height = baseline × multiplier + eye height

1. How tall is it? (tree, flagpole or house)

Object	Baseline (m)	Angle 1	Angle 2	Angle 3	Average angle	Multiplier	Eye height (m)	Height (m)

2. Practice

1. Baseline 25 m, angle 40°, eye height 1.3 m. Height = _____ m.
2. Baseline 80 m, angle 55°, eye height 1.5 m. Height = _____ m.
3. Scale drawing: on the back, draw a baseline of 10 squares (1 square = 2 m). Draw a 35° angle at one end and a line straight up at the other. How many squares tall? _____ = _____ m.

3. Track a stomp rocket

Launch	Tracker	Baseline (m)	Angle	Multiplier	Height (m)	Average of trackers (m)
1						
1						
2						
2						

4. Name two things that could make a height measurement wrong.

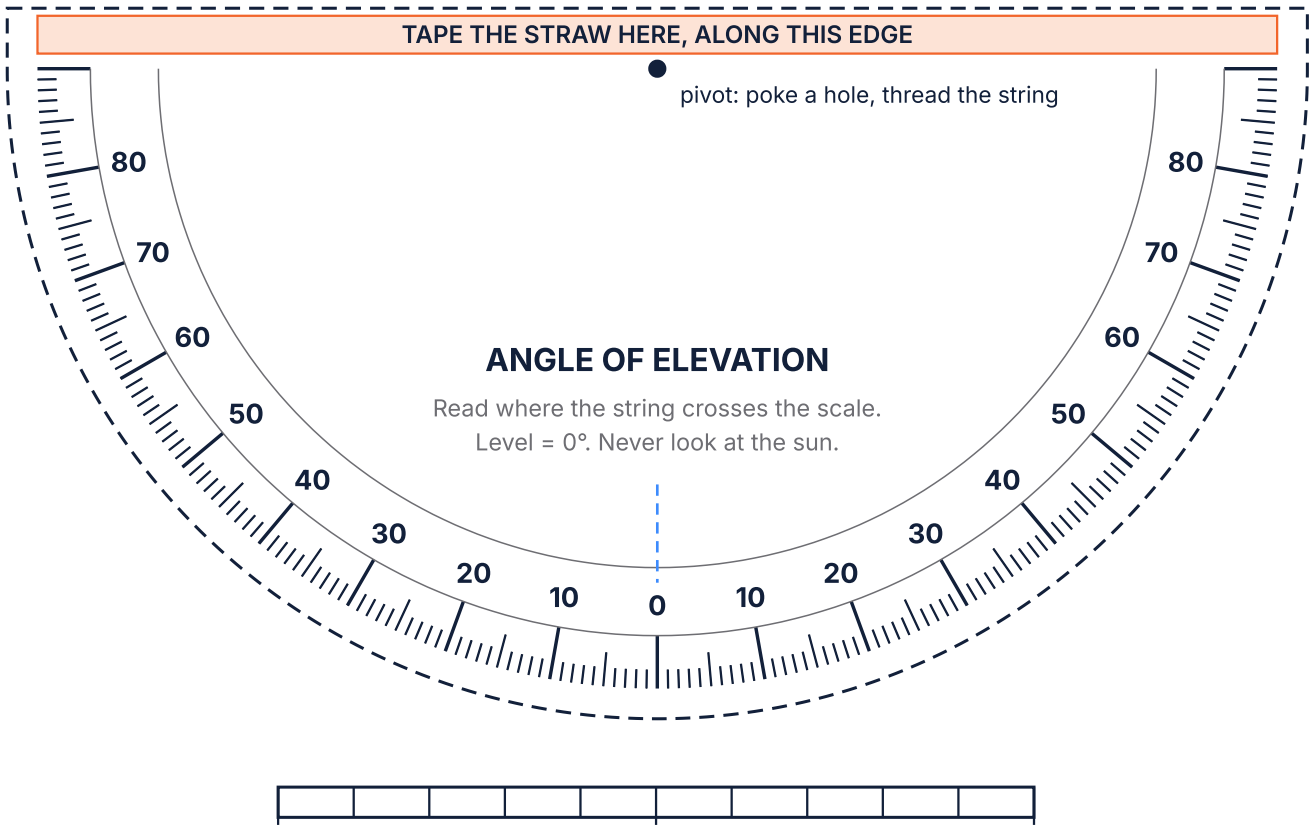
5. **Grades 7–8:** with a 50 m baseline, an angle of 45° gives _____ m and 47° gives _____ m. At 75°: _____ m; at 77°: _____ m. Which angle is more sensitive to a 2° mistake? _____

Inclinometer template

All ages

Day 8 · Print at 100% (actual size)

1. Print at **100%**. Check the 10 cm bar with a ruler. Glue onto cardstock and cut along the dashed line.
2. Tape a drinking straw along the top edge, on the orange band.
3. Poke a small hole at the pivot dot. Thread about 25 cm of string through it and tape the end at the back.
4. Tie a small weight (a washer or nut) to the string so it hangs below the scale.
5. Look through the straw at the top of the object. Pinch the string against the card, then read the angle.



This bar should measure exactly 10 cm. If not, print again at 100% ("actual size").

Angle	Multiplier	Angle	Multiplier
5°	0.09	45°	1.00
10°	0.18	50°	1.19
15°	0.27	55°	1.43
20°	0.36	60°	1.73
25°	0.47	65°	2.14
30°	0.58	70°	2.75
35°	0.70	75°	3.73
40°	0.84	80°	5.67

Check it: look through the straw at something at your own eye height. The string should hang on 0°.

Tip: make two, so two trackers can measure every launch from different sides.

Never look at the sun through the straw.

W9 Safety rules

All ages

Day 9 · Paper-rocket rules and the NAR Model Rocket Safety Code

Words: safety code · launch controller · safety key · misfire · LCO · RSO · preflight check

Paper-rocket rules (everyone): glasses on · never aim at anyone · wait for "launch!" · a grown-up runs fizzy rockets · never chase a rocket into a road, tree or power line.
Motor rockets (12+): grown-ups handle motors, igniters and the key · 15 ft back · count down from 5 · misfire: key out, wait 60 s · wind under 20 mph · never catch a rocket.

1. Is it safe?

Story	Safe or not safe?	Which rule?
Zoe puts on safety glasses before the stomp rocket launch.		
Max aims his straw rocket at his little sister.		
Lily drops the fizzy tablet in the canister herself and leans over it to watch.		
Sam's dad keeps the launch key in his pocket until everyone is at least 15 feet back.		
The motor rocket didn't launch, so Mia ran straight to the pad to look at it.		
It's a 25 mph wind, but the family really wants to fly today.		
The rocket landed in a tall tree, so Leo started climbing up to get it.		
Ava counts down from 5 out loud, and everyone watches the rocket.		
Uncle Jo wants a higher flight, so he tries to open the motor and add more fuel.		
Noah runs under the parachute to catch the rocket before it lands.		

2. Launch team jobs

Job	Who (grown-up or kid?)	Name
Countdown caller		
Stomper (paper rockets)		
Trackers (inclinometers)		
Recorder (launch log)		
Recovery crew		
12+: Launch Control Officer (key, button)		
12+: Range Safety Officer (checks the rocket)		
12+: loads the motor and igniter		

W9c Launch checklists

All ages

Day 9 · Get ready for launch day

1. Weather check (everyone)

Forecast wind: _____ mph. Under 20 mph? _____ (Less wind is better: under 10 mph means less drift.)
Rain? _____

2. Paper-rocket launch checklist (everyone)

- [ ] Safety glasses for everyone
- [ ] Stomp rocket, spare rockets and the launcher tube
- [ ] Straw rockets and straws
- [ ] Fizzy rocket supplies and water (the grown-up keeps the tablets)
- [ ] My best drop-test parachute
- [ ] Two inclinometers, a tape measure, two stopwatches
- [ ] Launch log on a clipboard, pencils
- [ ] Rocket Builder open on a tablet or phone, for ▶ Watch it launch
- [ ] Water, hats, sunscreen

The rest of this page is for the **12+ printed-rocket track**.

3. Field check (12+)

Field size: _____ m by _____ m. Rocket Builder's suggested field: _____ m. Big enough? _____ If not: a smaller motor that still gets a ✓, or a club launch.

4. Preflight checklist (12+)

- [ ] Weighed: _____ g (app said _____ g)
- [ ] Fins firm, no cracks
- [ ] Nose slides off smoothly
- [ ] Shock cord (braided flat elastic) tied tightly at both ends
- [ ] Parachute not torn, lines not tangled
- [ ] Balance point with motor (a grown-up puts it in, no igniter): _____ mm from the tip (app CG _____ mm)
- [ ] Swing test: nose leads
- [ ] Motor taken out again by a grown-up
- [ ] Launch box packed; motors, igniters and key with the grown-up

W9b Graph your week

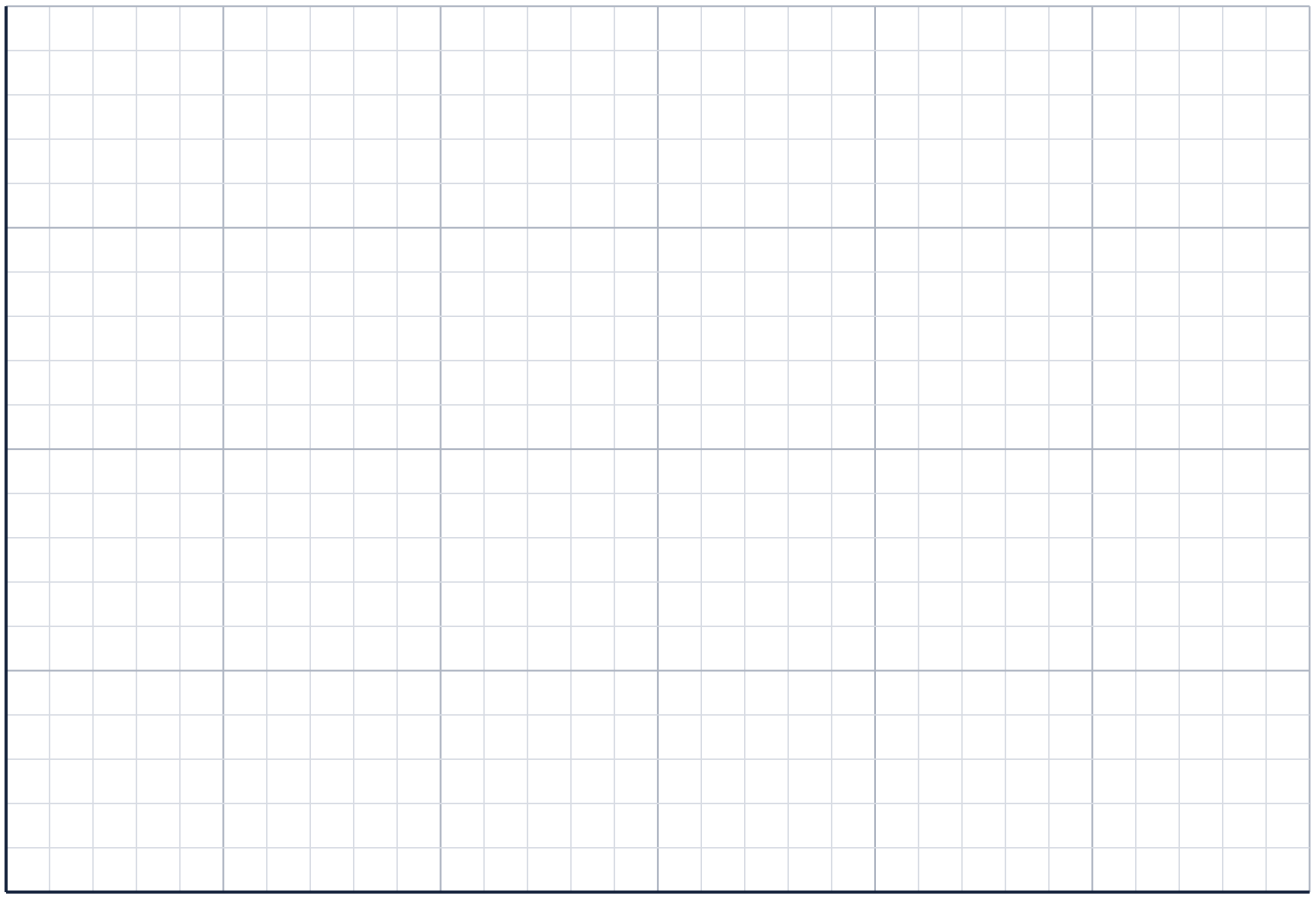
All ages

Day 9 · Data and graphing

My data set: _____ (from Day _____)

Graph type: bar graph / line graph / scatter plot (circle one). Remember: a title, labels on both axes, units, and evenly spaced numbers.

Title: _____



Mean of one group of trials: (_____ + _____ + _____) ÷ 3 = _____ **Range** = _____ − _____ = _____

1. As _____ went up, _____ went _____.
2. One thing that might have made my data less accurate is _____
3. **Grades 7–8:** draw a line of best fit by eye. Is the relationship a straight line? Where does it bend?

L Paper-rocket launch log

All ages

Day 10 · Stomp, straw and fizzy rockets

Date: _____ **Place:** _____ **Wind:** _____ mph **Tracker 1 baseline:** _____ m **Tracker 2 baseline:** _____ m

#	Rocket or launcher	What I changed	My guess (m)	Tracker 1 angle	Tracker 2 angle	Height (m)	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							

Height = baseline × multiplier + eye height (the multiplier table is on W8 and the template). Average the two trackers. Grades 3–4: write the angles and let a grown-up help with the multiplying.

1. Which launch went highest? What was different about it?

2. How close were your guesses? Did they get better as the day went on?

F Flight log (printed rocket)

12+ only

Day 10 · Launch day, 12+ track

	Flight 1	Flight 2	Flight 3
Rocket name			
Date and time			
Place			
Wind (mph) and direction			
Temperature			
Rocket mass, no motor (g)			
Motor (a grown-up loads it)			
Parachute size			
Predicted altitude (m)			
Tracker 1: baseline (m) and angle			
Tracker 2: baseline (m) and angle			
Measured altitude (m)			
Liftoff to parachute (s)			
Parachute to landing (s)			
Landed how far from the pad?			
Did it fly straight?			
Parachute opened near the top?			
Condition after landing			
Notes			

Altitude = baseline × multiplier (or × tan of the angle) + eye height. Average the trackers. Difference = predicted − measured.

Launch day rules: grown-ups handle motors, igniters and the key · glasses on, at least 15 ft back · count down from 5 · misfire: key out, wait 60 s · never chase a rocket into trees, roads or power lines · never catch it.

C Capstone planner and rubric

All ages

Day 10 · Mission report

Make a poster, slides, a written report or a 3–5 minute talk. Plan each part here.

1. Mission What did you set out to do?

2. Design Your Rocket Builder rocket with the blue and red dots and its score, and why you chose it (use Day 6 data). You can add your best paper rocket.

3. The science (a) Newton's 3rd law and thrust · (b) why the balance point is in front of the push point · (c) how a parachute works.

4. Guess vs launch Table or graph: your guesses and the heights you measured (12+: the app's prediction and the real flight).

5. What happened and why Reasons for any differences.

6. Next time One change, and what you predict it will do.

7. Safety Two rules your team followed and why they matter.

Rubric

	4 Excellent	3 Good	2 Getting there	1 Just starting	Score
Science	Explains Newton's 3rd law, stability and parachutes correctly, in own words, with examples	All three, small mistakes	One or two	Missing or mostly incorrect	
Design process	Several iterations with data; one change at a time; data justifies the choice	Iterations and a reason	A design, little data	No reasons	
Data and math	Launch log complete; heights correct; compared with the prediction (gr. 7–8: % difference)	Mostly complete; small errors	Some data; calculation incomplete	Little or no data	
Explaining results	Two or more sensible reasons; a testable next step	One reason and a next step	A reason or a next step	Neither	
Communication and safety	Clear, labelled pictures and graphs; two safety rules and why	Clear, one piece missing	Hard to follow in places	Most parts missing	

Total _____ / 20. 17–20 mission accomplished · 13–16 solid flight · 9–12 needs another iteration · below 9 back to the pad.

 Name _____ Date _____

Quick checks, Days 1–5

All ages

Answer on the line. Answers are in the day's lesson plan and in the answer keys.

Day 1

1. A rocket pushes gas down. The gas pushes the rocket up. Which of Newton's laws is this?
2. Two balloon rockets get the same push. One has 10 paper clips. Which speeds up more?
3. A rocket sits on the launch pad. Why doesn't it move until the motor fires?
4. True or false: a rocket needs air to push against.
5. Write F = m × a in words.

Day 2

1. Which force always points straight down?
2. Which force always points opposite to the motion?
3. During the coast, which forces act on the rocket?
4. Thrust is 6 N and weight is 1 N. Is thrust at least 5 times the weight?
5. Under a parachute the rocket falls at a steady speed. Balanced or unbalanced?

Day 3

1. Which dot must be in front: blue (balance point) or red (push point)?
2. Name two ways to move the push point backward.
3. Body 2 cm wide; push point 3 cm behind the balance point. Score?
4. Is a score of 0.5 cal OK to fly?
5. What happens to a 4 cal rocket on a windy day?

Day 4

1. Name three things that make drag bigger.
2. Twice as fast: how much bigger is the drag?
3. In a fair test, how many variables do you change at once?
4. Which has more drag: a flat nose or a pointed nose?
5. Why not put huge fins on every rocket?

Day 5

1. In B6-4, what do the B, the 6 and the 4 mean?
2. 5 N average for 2 s: total impulse? Class?
3. Which has more total push, C or B? About how much more?
4. Rocket with motor weighs 1.2 N; thrust 5 N. At least 5 times?
5. Who puts the igniter in the motor?

Q Quick checks, Days 6–10

All ages

Answer on the line. Answers are in the day's lesson plan and in the answer keys.

Day 6

1. Put in order: test, ask, create, improve, imagine, plan.
2. What's the difference between a criterion and a constraint?
3. Why change only one thing at a time?
4. Your design scores 0.8 cal. Name two things you could change.
5. Your rocket scores 3.6 cal. Is that a problem?

Day 7

1. A parachute falls 3 m in 1.5 s. Descent rate?
2. Bigger canopy, same weight: faster or slower?
3. At a steady speed, which two forces are balanced?
4. Rocket Builder aims for about what descent speed?
5. From 100 m at 4 m/s in a 4.5 m/s wind: how far does it drift?

Day 8

1. 30 m from a flagpole, looking up at 45°. How tall (no eye height)?
2. Baseline 50 m, angle 60° (multiplier 1.73). How high?
3. Which side of the triangle is the baseline?
4. Why add your eye height?
5. Should you ever look at the sun through the straw?

Day 9

1. How far back for a D motor or smaller?
2. After a misfire, what two things happen before anyone goes to the pad?
3. Highest wind speed you can launch in?
4. Who handles motors and igniters?
5. Trials 4 m, 5 m, 6 m: mean? range?

Day 10

1. Order: apogee, liftoff, burnout, landing, parachute opens.
2. What makes the rocket go up, using Newton's 3rd law?
3. Two trackers measured 12 m and 14 m. Average?
4. Predicted 150 m, flew 120 m. How many metres short? (Gr. 7–8: what percent?)
5. Who takes the spent motor out?

K Answer keys (1 of 3)

Grown-ups

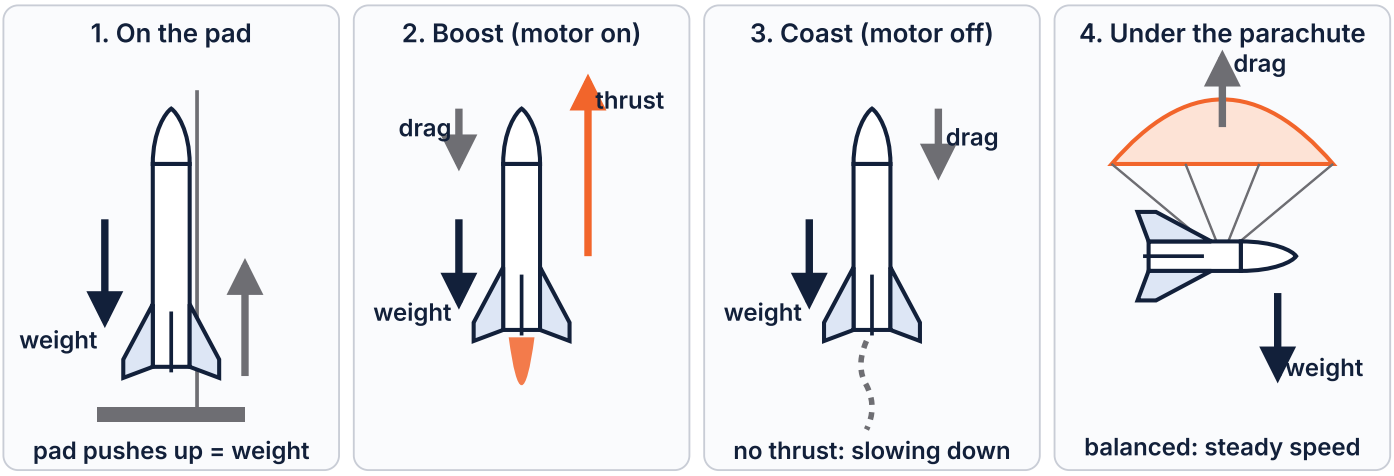
For grown-ups

W1 Balloon rocket

- Data varies. Expect: bigger balloons go further (more air = a longer push). More paper clips go less far (more mass, same push = less acceleration).
- Q1: the air pushes back on the balloon (action and reaction). Q2: it went less far / slower; the same force on more mass gives less acceleration. Q3: no unbalanced force until the air rushed out (law 1); friction on the string and air drag finally stopped it.
- Grades 7–8: 0.1 ÷ 0.010 = 10 m/s²; 0.1 ÷ 0.020 = 5 m/s².

W2 Forces

- 1 On the pad: weight down, the pad pushes up (equal). 2 Boost: big thrust up; weight and drag down. 3 Coast: weight and drag both down; no thrust. 4 Parachute: weight down, drag up (equal).
- Balanced: 1 and 4. Unbalanced: 2 and 3.
- Launch angle: about 40–45° usually goes furthest. Watch it launch order: liftoff 1, boost 2, coast 3, top 4, parachute 5. Power bar (12+): the app's thrust ÷ weight; it passes at 5× or more.



W3 Swing test

- Data varies. Usually C (fins + nose clay) is most stable: its balance point is furthest in front of its push point. A often tumbles; D is often wobbly.
- Nose clay moves the balance point forward (a bigger gap); tail clay moves it back, toward or behind the push point.
- 4.5 ÷ 3 = 1.5 cal: just right.

W4 Drag test

- Data varies. The flat nose usually goes least far (most drag). Cone and round are often close; small differences may be luck: more trials, a fairer launcher.
- Fin trade-off: no fins wobble and land short; medium fins fly straight; very large fins fly straight but may lose a little distance from drag and weight.
- Grades 7–8: 0.5 × 1.2 × 1600 × 0.65 × 0.00071 ≈ 0.44 N. Lower Cd gives a higher predicted altitude.

K Answer keys (2 of 3)

Grown-ups

For grown-ups

W5 Motor codes and impulse

- Decode: A8-3 = A, 8 N, 3 s, 30 m · B6-4 = B, 6 N, 4 s, 60 m · C6-5 = C, 6 N, 5 s, 120 m · D12-5 = D, 12 N, 5 s, 150 m.
- Impulse: a 3 × 0.8 = 2.4 N·s, A · b 5 × 0.9 = 4.5 N·s, B · c 10 × 1.7 = 17 N·s, D · d 15 × 3.4 = 51 N·s, F.
- Thrust-to-weight: 0.8 N; 5 ÷ 0.8 ≈ 6.3; pass. 1.2 N; 4.7 ÷ 1.2 ≈ 3.9; does not pass (too heavy for this motor).
- Part 3: the motor and height come from the app's How high will it go? card. Fizzy rockets: data varies; look for a sensible reason (room for gas to build pressure vs. enough water for the tablet to fizz fast).

W5b Thrust curve

- Peak 12 N at about 0.05 s.
- Area ≈ 0.3 + 0.54 + 0.73 + 0.45 + 2.2 + 0.1 ≈ 4.3 N·s: a B motor (2.51–5).
- Average ≈ 4.3 ÷ 0.86 ≈ 5.0 N. The code's number is rounded, and each real motor differs a little.
- Weight ≈ 0.84 N; net ≈ 3.9 N; a ≈ 3.9 ÷ 0.086 ≈ 45 m/s² (about 4½ g).

W6 Design log

- Data varies with the app version and choices. Typical trend: with the same fins, taller rockets score higher. Very short rockets show “fins grew” and a score just above 1.
- Sample (M18, standard fins, body 350 mm, about 400 mm tall): clipped delta 2.0, trapezoid 1.7, swept 1.7, delta 1.4, rectangle 0.7 cal (the app would enlarge the rectangle's fins). Swept-back fins move the push point further back.
- Choice: any design meeting the criteria, with a reason drawn from the data.

W7 Parachute drop test

- Data varies. Bigger canopies fall slower; more washers fall faster; no parachute lands in well under a second.
- Practice: 120 ÷ 4 = 30 s; 30 × 4.5 = 135 m. A 60 m field is not big enough for the drift.
- Too big a parachute: the rocket hangs in the air longer and drifts further (trees, roads, outside the field).

W8 Altitude tracking

- Practice 1: 25 × 0.84 + 1.3 = 22.3 m. Practice 2: 80 × 1.43 + 1.5 ≈ 116 m. Scale drawing: about 7 squares = 14 m (tan 35° × 20 m = 14 m).
- Errors: angle reading (string swinging), baseline measured wrong, not catching the very top, the rocket drifting toward or away from you, eye height forgotten.
- Grades 7–8: 50.0 m and 53.6 m; 186.6 m and 216.6 m. Steep angles (75°) are much more sensitive.

K Answer keys (3 of 3)

Grown-ups

For grown-ups

W9 Safety rules

- Stories: 1 safe (glasses on) · 2 not safe (never aim at anyone) · 3 not safe (a grown-up handles the tablets; never lean over a canister) · 4 safe (key with the grown-up; 15 ft back) · 5 not safe (misfire: key out, wait 60 s; grown-ups handle motors) · 6 not safe (no launches over 20 mph) · 7 not safe (never recover from tall trees) · 8 safe (count down out loud) · 9 not safe (never open or change a motor) · 10 not safe (never catch a rocket).
- Jobs: countdown caller, stomper, trackers, recorder and recovery crew = kids (recovery only when the grown-up says so). LCO, RSO and loading motors/igniters = grown-ups (12+ track).

W9b Graph your week

- Check: title, both axes labelled with units, even scale, correct plotting, a sensible trend sentence. Grades 7–8: drop-test speed goes with 1 ÷ √area, so it curves rather than falling in a straight line.

Quick checks

Day 1: 1. Third law. 2. The one with no paper clips. 3. Forces are balanced; it stays put until thrust (law 1). 4. False. 5. Force equals mass times acceleration.

Day 2: 1. Weight. 2. Drag. 3. Weight and drag (no thrust). 4. Yes: 6 ÷ 1 = 6. 5. Balanced.

Day 3: 1. Blue (balance point). 2. Bigger fins, swept-back fins, a longer body. 3. 1.5 cal. 4. No: wobbly, can't download. 5. It weathercocks into the wind.

Day 4: 1. Speed, frontal area, blunt shape, roughness. 2. About 4 times. 3. One. 4. Flat nose. 5. Extra drag and weight; can be over-stable.

Day 5: 1. Class B, about 6 N average thrust, 4 s delay. 2. 10 N·s, class C. 3. C, up to about twice. 4. No: about 4.2. 5. A grown-up.

Day 6: 1. Ask, imagine, plan, create, test, improve. 2. Criterion = what it must do; constraint = a limit. 3. To know which change caused the result. 4. Taller, bigger fins, lighter motor, nose weight. 5. Yes, over-stable: weathercocks.

Day 7: 1. 2 m/s. 2. Slower. 3. Drag and weight. 4. About 4.5 m/s or less (3–5 m/s). 5. 25 s × 4.5 ≈ 113 m.

Day 8: 1. 30 m. 2. About 87 m (+ eye height). 3. Along the ground, you to the base. 4. The angle is measured from your eye. 5. Never.

Day 9: 1. 15 feet (5 m). 2. Key out, wait 60 seconds. 3. 20 mph. 4. A grown-up. 5. Mean 5 m; range 2 m.

Day 10: 1. Liftoff, burnout, apogee, parachute opens, landing. 2. Motor pushes gas down; gas pushes rocket up (stomp: air pushes it off). 3. 13 m. 4. 30 m; 20%. 5. A grown-up, after it cools.

Flight log and capstone

- Use the rubric. Check the altitude math: baseline × multiplier + eye height, averaged over trackers; percent difference = (predicted − measured) ÷ predicted × 100.